

N A Adarsh Pritam

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OVERVIEW

I am currently a **Research Intern at the Centre for Brain Research (CBR), Indian Institute of Science (IISc)**, where I work on **machine learning and statistical methods for dementia research** (for early detection of cognitive decline). I recently completed my MSc in Data Science from Alliance University, where my thesis focused on **brain-to-latent representation learning and generative modeling for biomedical imaging**. Previously, I was a **Research Intern at the i2CS Research Group, Indian Institute of Information Technology (IIIT) Kottayam**, where I developed **vision-language models for plant disease diagnosis**. My research interests include **Representation Learning, Generative Models, Computer Vision, Multimodal Learning, and Medical AI**.

EDUCATION

2024 – 2026	Master of Science, Data Science Machine Learning I & II, Deep Learning, ML for Image Processing, Natural Language Processing, MLOps, Data Science, Research Methodology.	Alliance University Bangalore, India (Sem 4 SGPA: 9.7/10)
2021 – 2024	Bachelor of Science, Mathematics and Statistics Mathematical Statistics, Probability, Applied Mathematics.	Bengaluru City University Bangalore, India (CGPA: 6.5/10)

RELEVANT EXPERIENCE

CBRAIN Research Intern **May 2026 – Present**
Centre for Brain Research (CBR) @ Indian Institute of Science (IISc), Bangalore
Principal Investigator: Dr. Thomas Gregor Issac

- Researching the applications of Multimodal Machine Learning in Dementia Research for the early detection of Mild Cognitive Impairment using longitudinal clinical, cognitive, biochemical, and neuroimaging data from the ongoing flagship studies TLSA and SANSCOG.

Graduate Research Student **Aug 2025 – May 2026**
Alliance University, Bangalore
Supervisor: Dr. Jeba Shiney

- Developed **Brain2VLM**, for brain-to-image reconstruction, decoding fMRI activity into diffusion latents and CLIP embeddings, revealing linear vs nonlinear alignment regimes ($\Delta \approx 0.47$) and improving reconstruction performance (CLIP $\approx 85\%$).
- Developed **SkinGenBench**, a generative benchmark on dermoscopic data (14K+ images), showing GAN-based models outperform diffusion, achieving ROC-AUC 0.98 and 8–15% gains.

Summer ML Research Intern **May 2025 – Sep 2025**
i2CS Research Group @ Indian Institute of Information Technology (IIIT) Kottayam
Supervisor: Dr. Kala S

- Developed multimodal plant disease detection system using CLIP (ViT-L) + InternLM2. Designed projection module for aligning vision and language representations. Implemented scalable training pipelines and contributed to research manuscript.

Glimpse of MSc. Thesis

Title: Learning Structured Representations: From Brain-to-Latent Alignment to Generative Modeling in Biomedical Imaging

Supervisor: Prof Jeba Shiney

Abstract: Investigates hierarchical brain-to-latent alignment and generative modeling in two complementary works.

(1) **Brain2VLM** studies hierarchical brain-to-latent alignment by decoding fMRI activity into diffusion latents and CLIP embedding spaces using linear (ridge) and nonlinear (residual MLP) decoders. On the Natural Scenes Dataset, results show that linear mappings suffice for structural latents ($\Delta \approx 0.05$ – 0.06), while nonlinear decoding substantially improves semantic alignment ($\Delta \approx 0.47$, MMD 0.042 vs 0.358), leading to stronger reconstruction performance (PixCorr ≈ 0.33 , CLIP $\approx 85\%$) and exposing a trade-off between alignment accuracy and generative compatibility. Preprint: <https://www.biorxiv.org/content/10.64898/2026.04.23.720313v1>, Code: <https://github.com/adarsh-crafts/Brain2VLM>

(2) **SkinGenBench** introduces a systematic biomedical imaging benchmark analyzing the interplay between preprocessing complexity and generative model choice for synthetic dermoscopic augmentation. Across 14,116 images from HAM10000 and MILK10K, we show that generative architecture dominates preprocessing, with StyleGAN2-ADA achieving superior fidelity (FID ≈ 65.5 , KID ≈ 0.05) and delivering 8–15% gains in melanoma F1-score (ViT-B/16 F1 ≈ 0.88 , ROC-AUC ≈ 0.98). Preprint: <https://arxiv.org/abs/2512.17585>, Code: <https://github.com/adarsh-crafts/SkinGenBench>

ONGOING RESEARCH PAPERS AND PROJECTS

- **List of Publications:** <https://scholar.google.com/citations?hl=en&user=eQ6t2UkAAAAJ>
- **(ongoing, preprint)** Brain2VLM: Hierarchical Alignment Between Cortical Representations and Vision-Language Latent Spaces. Available at: <https://www.biorxiv.org/content/10.64898/2026.04.23.720313v1>
- **(Under review)** SkinGenBench: Generative Model and Preprocessing Effects for Synthetic Dermoscopic Augmentation in Melanoma Diagnosis. Available at: <https://arxiv.org/abs/2512.17585>
- **IEEE - ETCC 2025:** American-Sign-Language to Fluent-English Detection and Translation Using T5. Available at: <https://ieeexplore.ieee.org/document/11108641/>
- **Project LLaMA LLM Reimplementation:** Built an LLM from scratch to mimic communication style using personal chat logs. Implemented tokenizer, transformer, and QLoRA-based fine-tuning. Available at: <https://github.com/adarsh-crafts/llama-llm-from-scratch> and [Medium Blog](#)
- **(Open-Source Contribution) PyTorch Classification Extended:** Extended the bearpaw PyTorch image classification framework with Grad-CAM visualization and expanded support to more architectures. Available at: <https://github.com/adarsh-crafts/pytorch-classification-extended>
- **Project Life Expectancy Modeling:** Built an end-to-end ML pipeline on WHO data to model global life expectancy, incorporating preprocessing, multicollinearity analysis, and interpretable regression (Linear Regression, Decision Trees) with standard evaluation. deployed via Streamlit for interactive analysis. <https://github.com/adarsh-crafts/life-expectancy-modeling>

SKILLS

Core: Deep Learning, Multimodal Learning, Generative Models, Representation Learning

Frameworks: PyTorch, Transformers, OpenCV, Scikit-learn

Engineering: Docker, MLFlow, AWS, Git

Data: NumPy, pandas, SQL

Languages: English, Telugu, Hindi

Hobbies: Guitar, Trekking

CERTIFICATIONS & COURSES

- AIF-C01: AWS Certified AI Practitioner (2024)
- Ethics in Data Science – University of Michigan (Coursera) (2026)
- The Data Scientist’s Toolbox – John Hopkins University (Coursera) (2025)
- NVIDIA Fundamentals of Deep Learning (2025)
- Mathematics for Machine Learning and Data Science – DeepLearning.AI (2024)
- Google Data Business Intelligence Specialization (2023)
- Google Data Analytics Professional Certificate (2023)

AWARDS & ACHIEVEMENTS

- Invited to deliver seminars on machine learning research methodology for MSc Data Science students.
- Project Lead for Academic Review Papers and other projects for courseworks
- Won multiple Battle of the bands (Musical Competitions) in and outside the university.
- High Achiever’s Award, Distinction in Grade 4 Rock Guitar Examinations (London College of Music)

REFERENCES

1. Dr. Thomas Gregor Issac (thomasgregor@cbr-iisc.ac.in), Associate Professor, Centre for Brain Research, Indian Institute of Science.
2. Dr. Jeba Shiney (jeba.shiney@alliance.edu.in), Professor, Alliance School of Advanced Computing, Alliance University.
3. Dr. Vijayalakshmi Nanjappan (vijayalakshmi.n@alliance.edu.in), Associate Professor, Alliance School of Advanced Computing, Alliance University.
4. Dr. Kala S (kala@iiitkottayam.ac.in), Assistant Professor & Head of i2CS Research Group, Indian Institute of Information Technology (IIIT) Kottayam.